

Supplementary Material for Physical-Support Confidence Sets for Highly Coherent Dictionaries

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1 Proofs and Technical Details

The supplement is organized by theorem dependency. Technical lemmas are stated immediately before the main proof steps that use them, so that the statistical argument can be read without moving between separate proof and technical-detail files.

1.1 Calibration geometry and orientation information

1.1.1 Projector chart and exchangeable subset aggregation

For $z \in U$, write

$$d(z) = c(z)u + z, \quad c(z) = \sqrt{1 - \|z\|^2},$$

and define

$$H(z) = P\{d(z)\} - P_u.$$

Using

$$c(z) = 1 - \frac{1}{2}\|z\|^2 + O(\|z\|^4),$$

the projector increment admits the total-degree expansion

$$H(z) = A_1(z) + A_2(z) + A_3(z) + R_4(z), \quad (1)$$

with

$$A_1(z) = uz^\top + zu^\top, \quad (2)$$

$$A_2(z) = zz^\top - \|z\|^2 P_u, \quad (3)$$

$$A_3(z) = -\frac{1}{2}\|z\|^2(uz^\top + zu^\top), \quad (4)$$

and, uniformly on the enlarged local cap,

$$\|R_4(z)\|_F \leq C\|z\|^4, \quad \|DR_4(z)\|_{\text{op}} \leq C\|z\|^3. \quad (5)$$

Fix a stratum indexed by support size k and separated-atom indicator A , let $h = k - A$, and define

$$C_0 = \nu I_q + \lambda_k(hP_u + AP_a).$$

Only strata with $0 \leq h \leq q$ and positive weight are retained. For a coherent-atom subset C , put

$$X_m(C) = \lambda_k \sum_{j \in C} A_m(z_j), \quad m = 1, 2, 3.$$

Let $\mathcal{D}_m = D^m \gamma_{C_0}$ denote the m th covariance derivative of the Gaussian density at C_0 . Retaining monomials by total degree in the coherent-atom coordinates gives

$$\mathcal{P}_{0,C} = \gamma_{C_0}, \quad (6)$$

$$\mathcal{P}_{1,C} = \mathcal{D}_1[X_1(C)], \quad (7)$$

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$$\mathcal{P}_{2,C} = \mathcal{D}_1[X_2(C)] + \frac{1}{2}\mathcal{D}_2[X_1(C), X_1(C)], \quad (8)$$

$$\mathcal{P}_{3,C} = \mathcal{D}_1[X_3(C)] + \mathcal{D}_2[X_1(C), X_2(C)] + \frac{1}{6}\mathcal{D}_3[X_1(C), X_1(C), X_1(C)]. \quad (9)$$

Every omitted term has total coherent-atom degree at least four.

The cancellations responsible for the cubic orientation effect appear only after averaging over exchangeable subsets. Let $H_1, \dots, H_q$ belong to a commutative symmetric tensor algebra, define

$$Q_m = \sum_{j=1}^q H_j^{\odot m}, \quad X_C = \sum_{j \in C} H_j, \quad \pi_m(h) = \frac{(h)_m}{(q)_m},$$

and let C be uniformly distributed over the h -subsets of $[q]$. Partitioning ordered index tuples by equality pattern yields

$$\mathbb{E}_h X_C = \pi_1 Q_1, \quad (10)$$

$$\mathbb{E}_h X_C^{\odot 2} = (\pi_1 - \pi_2) Q_2 + \pi_2 Q_1^{\odot 2}, \quad (11)$$

$$\mathbb{E}_h X_C^{\odot 3} = (\pi_1 - 3\pi_2 + 2\pi_3) Q_3 + 3(\pi_2 - \pi_3) Q_1 \odot Q_2 + \pi_3 Q_1^{\odot 3}. \quad (12)$$

For the projector expansion, every degree-at-most-three subset aggregate reduces to the coherent-atom power sums

$$B_1 = \sum_j z_j, \quad B_2 = \sum_j z_j^{\otimes 2}, \quad B_3 = \sum_j z_j^{\otimes 3}, \quad (13)$$

and their products. The exact affine-simplex identities are

$$B_1 = qb, \quad (14)$$

$$B_2 = qb^{\otimes 2} + \frac{q}{q-1} G_2, \quad (15)$$

$$B_3 = qb^{\otimes 3} + \frac{q}{q-1} \text{Sym}_3(b \otimes G_2) + G_3. \quad (16)$$

Hence no residual orientation coordinate appears at degree one or two; orientation first enters through the cubic term $G_3 = L^{\otimes 3} T_q$.

1.1.2 Gaussian derivative envelope and composed remainder

The following envelope controls all Taylor terms used in the calibration density expansion.

Lemma 1.1 (Uniform Gaussian derivative envelope). *Suppose the eigenvalues of positive-definite A lie in a fixed compact subset of $(0, \infty)$ and*

$$\|A^{-1/2}(B - A)A^{-1/2}\|_{\text{op}} \leq \rho < \frac{1}{2}.$$

Then, for $0 \leq m \leq 5$,

$$\int \frac{|D^m \gamma_A[H_1, \dots, H_m]|^2}{\gamma_B} \leq C_m \prod_{j=1}^m \|H_j\|_F^2. \quad (17)$$

Proof. Each covariance derivative is a polynomial of degree $2m$ in the observation times γ_A . For example,

$$D\gamma_A[H](y) = \frac{\gamma_A(y)}{2} \{y^\top A^{-1} H A^{-1} y - \text{tr}(A^{-1} H)\}.$$

Moreover,

$$2A^{-1} - B^{-1} \succeq \left(2 - \frac{1}{1-\rho}\right) A^{-1} \succ 0,$$

so the required polynomial moments of the Gaussian quotient are uniformly finite. $\square$

Let $\mathcal{G}_{k,A}$ denote the already support-aggregated stratum density, let $\mathcal{P}_{\leq 3,k,A}$ be its total-degree-three polynomial, and write

$$R_{\geq 4,k,A} = \mathcal{G}_{k,A} - \mathcal{P}_{\leq 3,k,A}.$$

Integral Taylor formulas and Lemma 1.1 give

$$\left\| \frac{D_z R_{\geq 4, k, A}[\dot{z}]}{\sqrt{q_{k, A, \theta'}}} \right\|_2 \leq C s^3 \max_j \|\dot{z}_j\|, \quad (18)$$

$$\left\| \frac{D_{(P_a, \nu)} R_{\geq 4, k, A}[\dot{P}_a, \dot{\nu}]}{\sqrt{q_{k, A, \theta'}}} \right\|_2 \leq C s^4 (\|\dot{P}_a\|_F + |\dot{\nu}|). \quad (19)$$

Mixture aggregation introduces no inverse minimum-weight factor because, for nonnegative weights w_ℓ ,

$$\frac{(\sum_\ell w_\ell r_\ell)^2}{\sum_\ell w_\ell q_\ell} \leq \sum_\ell w_\ell \frac{r_\ell^2}{q_\ell}. \quad (20)$$

After optimal dictionary alignment,

$$s^3 \max_j \|\Delta z_j\| \leq C \{s^3 \|\Delta b\| + s^2 \|\Delta G_2\| + s \|\Delta G_3\|\} \leq C s \delta_F. \quad (21)$$

Integrating the derivative bounds therefore yields

$$\left\| \frac{R_{\geq 4}(\theta) - R_{\geq 4}(\theta')}{\sqrt{f_{\theta'}}} \right\|_2 \leq C s \delta_F(\theta, \theta'). \quad (22)$$

1.1.3 Simplex cubic and quotient inverse

For unit $x \in U$, set $y_j = v_j^\top x$. Then

$$\sum_j y_j = 0, \quad \sum_j y_j^2 = \frac{q}{q-1}, \quad T_q[x, x, x] = \sum_j y_j^3. \quad (23)$$

The tight-frame map

$$x \mapsto (v_1^\top x, \dots, v_q^\top x)$$

is a linear isomorphism from U onto the zero-sum hyperplane. Optimizing $\sum_j y_j^3$ under the constraints in Equation (23) therefore reduces to an exact finite-dimensional constrained problem. At a stationary point the coordinates take at most two values; if k coordinates take the positive value, the objective is

$$\frac{q(q-2k)}{(q-1)^{3/2} \sqrt{k(q-k)}}. \quad (24)$$

This is maximized at $k = 1$, corresponding exactly to a simplex vertex. Hence

$$\text{Stab}_{O(U)}(T_q) = \mathfrak{S}_q. \quad (25)$$

The orbit differential satisfies

$$\|\Omega \cdot T_q\|_F^2 = 3 \left(\frac{q}{q-1} \right)^3 \|\Omega\|_F^2. \quad (26)$$

Consequently, with

$$d_{\text{orb}}([R], [R']) = \min_{\pi \in \mathfrak{S}_q} \|R - R' \Pi_\pi\|_F,$$

the constant-rank theorem locally and compactness globally give

$$d_{\text{orb}}([R], [R']) \leq C \|R^{\otimes 3} T_q - R'^{\otimes 3} T_q\|_F. \quad (27)$$

For a general shell point, write the left polar decomposition

$$L = P_L O_L, \quad P_L = (L L^\top)^{1/2},$$

and normalize the cubic invariant by

$$W(L) = P_L^{-\otimes 3} G_3.$$

Square-root functional calculus gives

$$\|P_L - P'_L\|_F \leq C s^{-1} \|G_2 - G'_2\|_F,$$

while

$$\|W(L) - W(L')\|_F \leq C [s^{-3} \|G_3 - G'_3\|_F + s^{-2} \|G_2 - G'_2\|_F]. \quad (28)$$

Together with Equation (27), these estimates yield the physical quotient inverse in Equation (53).

1.1.4 Collapsed moment operator and all-pair statistical chords

Use the separated-atom chart

$$a(w) = \frac{a_0 + w}{\|a_0 + w\|}, \quad w \in a_0^\perp,$$

and write

$$\Xi = (B_1, B_2, B_3, w, \nu).$$

For $X = tu + x$, $x \in U$, define

$$S_{\dot{B}}(t, x) = 2t\dot{B}_1[x] + \dot{B}_2[x, x] - t^2 \text{tr } \dot{B}_2 - t\dot{B}_3[x, I_U], \quad (29)$$

$$Q_{\dot{B}}(t, x) = 4t^3\dot{B}_1[x] + 6t^2\dot{B}_2[x, x] - 2t^4 \text{tr } \dot{B}_2 \quad (30)$$

$$+ 4t\dot{B}_3[x, x, x] - 6t^3\dot{B}_3[x, I_U]. \quad (31)$$

Let

$$H(\dot{w}) = a_0\dot{w}^\top + \dot{w}a_0^\top, \quad h_0(t, x) = \frac{1}{2}(t + e_1^\top x)^2,$$

and define

$$\mathcal{A}_q(\dot{B}, \dot{w}, \dot{\nu}) = (\dot{\nu}, S_{\dot{B}} + H(\dot{w})[X, X], Q_{\dot{B}} + 2h_0H(\dot{w})[X, X]). \quad (32)$$

Lemma 1.2 (Injectivity of the collapsed moment operator). *For every fixed $q \geq 3$, $\mathcal{A}_q$ is injective. Consequently there exists $\sigma_q > 0$ such that*

$$\|\mathcal{A}_q h\| \geq \sigma_q \|h\|. \quad (33)$$

Proof. If $\mathcal{A}_q(\dot{B}, \dot{w}, \dot{\nu}) = 0$, then $\dot{\nu} = 0$ and

$$H(\dot{w})[X, X] = -S_{\dot{B}}(t, x).$$

Substitution into the fourth-order output gives

$$Q_{\dot{B}}(t, x) = (t + e_1^\top x)^2 S_{\dot{B}}(t, x).$$

Comparing the coefficients of t^4, t^0, t^2, t^3, t^1 successively yields

$$\text{tr } \dot{B}_2 = 0, \quad \dot{B}_2 = 0, \quad 2\dot{B}_1 - \dot{B}_3[\cdot, I_U] = 0,$$

then $\dot{B}_1 = 0$, $\dot{B}_3 = 0$, and finally $H(\dot{w}) = 0$. Since the separated-atom tangent map is injective on $a_0^\perp$, $\dot{w} = 0$. $\square$

The nonlinear moment map

$$\mathcal{R}(\theta) = (\nu, \Sigma_2, \Sigma_4)$$

satisfies, after quotient alignment,

$$\mathcal{R}(\theta) - \mathcal{R}(\theta') = \mathcal{A}_q\{\Xi(\theta) - \Xi(\theta')\} + \text{Rem}_q(\theta, \theta'), \quad (34)$$

with

$$\|\text{Rem}_q(\theta, \theta')\| \leq C_q s \|\Xi(\theta) - \Xi(\theta')\|. \quad (35)$$

Choosing the supplied s_0 sufficiently small yields the uniform same-shell moment chord

$$c\delta_F(\theta, \theta') \leq \|\Delta(\nu, \Sigma_2, \Sigma_4)\| \leq C\delta_F(\theta, \theta'). \quad (36)$$

For a fixed exchangeable law, write

$$f_\theta = \sum_{I: w_I > 0} w_I \gamma_{C_I(\theta)}.$$

The Gaussian envelope and remainder bounds imply, locally in the noise coordinate and in both ordered directions,

$$\chi^2(f_\theta \| f_{\theta'}) \leq C\delta_F^2(\theta, \theta').$$

For a fixed separation in ν , compactness and log-sum absorb the finite componentwise KL bound into δ_F^2 . For the reverse Hellinger bound, use

$$\Psi(Y) = \{\text{vech}(YY^\top), \text{symvec}(Y^{\otimes 4})\}.$$

Uniform eighth moments imply

$$\|\mathbb{E}_\theta \Psi - \mathbb{E}_{\theta'} \Psi\| \leq CH(f_\theta, f_{\theta'}),$$

and the explicit moment inverse together with Equation (36) gives

$$\delta_F \leq CH(f_\theta, f_{\theta'}).$$

This proves the all-pair fixed-law chord in Equation (51).

For Bernoulli coding,

$$w_I(p) = p^{|I|}(1-p)^{n-|I|}.$$

The compact branch $p \in [p_-, p_+]$ gives

$$\sum_I \frac{\{w_I(p) - w_I(p')\}^2}{w_I(p')} \leq C|p - p'|^2,$$

while Equations (31) and (36) give the reverse Lipschitz control. Combining the weight and fixed-law changes proves Equation (52).

1.1.5 Point-adapted nuisance geometry and quadratic-mean differentiability

At an interior shell point write

$$L = sRG, \quad R \in O(U), \quad G = G^\top > 0,$$

and use the point-adapted path

$$L_t = R(sG + tS)e^{t\Omega}, \quad \dot{L} = R(S + sG\Omega), \quad S^\top = S, \quad \Omega^\top = -\Omega. \quad (37)$$

Together with local Euclidean paths in (p, ν, b) and the separated-atom projector tangent, write

$$h = (\dot{p}, \dot{\nu}, \dot{b}, \dot{P}_a, S, \Omega),$$

with $\dot{p}$ omitted for a fixed known law.

Direct differentiation gives

$$DG_2[h] = sR(SG + GS)R^\top, \quad (38)$$

$$DG_3[h] = \mathcal{S}_{R,G,s}[S] + (sRG)^{\otimes 3}(\Omega \cdot T_q), \quad \|\mathcal{S}_{R,G,s}[S]\|_F \leq Cs^2\|S\|_F. \quad (39)$$

The Lyapunov map $S \mapsto SG + GS$ is uniformly coercive on the normalized compact shape class, and Equation (26) controls the orientation component. Therefore

$$\|DF_p(\theta)[h]\|^2 \asymp |\dot{p}|^2 + |\dot{\nu}|^2 + \|\dot{b}\|^2 + \|\dot{P}_a\|_F^2 + s^2\|S\|_F^2 + s^6\|\Omega\|_F^2. \quad (40)$$

Let

$$\ell_{\theta,h}(y) = \frac{Df_\theta[h](y)}{f_\theta(y)}, \quad I_\theta(h) = \mathbb{E}_\theta \ell_{\theta,h}^2.$$

The Gaussian derivative envelope gives the upper Fisher bound, while the score-moment identity

$$D\mathbb{E}_\theta \Psi[h] = \mathbb{E}_\theta \{(\Psi - \mathbb{E}_\theta \Psi)\ell_{\theta,h}\}$$

and the differentiated moment chord give the reverse bound. Hence

$$c\|h\|_{\text{mix},\theta}^2 \leq I_\theta(h) \leq C\|h\|_{\text{mix},\theta}^2, \quad (41)$$

where the mixed norm is the right side of Equation (40).

On every compact interior subbranch, the finite Gaussian mixture is quadratic-mean differentiable in this chart. Indeed, the component weights and covariance matrices are twice continuously differentiable, the covariance spectra are uniformly bounded above and away from zero, and the Gaussian envelope controls first and second chart derivatives in $L^2(f_\theta^{-1})$. Taylor's theorem therefore gives

$$\left\| \sqrt{f_{\theta+u}} - \sqrt{f_\theta} - \frac{1}{2} \frac{Df_\theta[u]}{\sqrt{f_\theta}} \right\|_2 = o(\|u\|).$$

Define the nuisance-profiled orientation information by

$$I_{R|\eta,\theta}(\Omega) = \inf_{(\dot{p}, \dot{\nu}, \dot{b}, \dot{P}_a, S)} I_\theta(\dot{p}, \dot{\nu}, \dot{b}, \dot{P}_a, S, \Omega). \quad (42)$$

Taking the infimum in the lower half of Equation (41), and setting nuisance tangents to zero for the upper half, gives

$$cs^6\|\Omega\|_F^2 \leq I_{R|\eta,\theta}(\Omega) \leq Cs^6\|\Omega\|_F^2, \quad (43)$$

which establishes Equation (55).

1.1.6 Proof of the calibration-geometry results

Proof of Lemma 2.2 and Theorem 3.1. Conditional on the active set, one calibration observation is Gaussian with covariance

$$C_I = \nu I_q + \lambda_K \sum_{j=1}^n I_j P_j.$$

Exchangeability gives, for $i \neq j$,

$$\mathbb{E}(\lambda_K I_j) = a_1, \quad \mathbb{E}(\lambda_K^2 I_j) = a_{2d}, \quad \mathbb{E}(\lambda_K^2 I_i I_j) = a_{2o}. \quad (44)$$

Therefore

$$M_2 = \nu I_q + a_1 \Sigma_2(D)$$

and, by the conditional Gaussian fourth-moment identity,

$$K_4 = 3 \{ \mathbb{E}(C_I^{\odot 2}) - (\mathbb{E}C_I)^{\odot 2} \}.$$

Separating diagonal and off-diagonal atom pairs yields

$$\mathbb{E} \left(\lambda_K \sum_j I_j P_j \right)^{\odot 2} = a_{2o} \Sigma_2^{\odot 2} + (a_{2d} - a_{2o}) \Sigma_4,$$

so Equations (34)–(35) hold and

$$\Delta_{\mathcal{L}} = a_{2d} - a_{2o} = \frac{\mathbb{E}\{\lambda_K^2 K(n-K)\}}{n(n-1)}. \quad (45)$$

Under Equation (11), the resulting population-moment map is uniformly invertible on the compact known-law class. For Bernoulli–Gaussian coding, $a_1 = a_{2d} = p$ and $a_{2o} = p^2$, and Equations (31)–(36) give the stated identification of p , ν , Σ_2 , and Σ_4 . This proves Lemma 2.2.

The remaining claims follow from the technical results immediately above. The projector expansion and exchangeable subset aggregation show that all orientation dependence through degree two cancels after support averaging, while the first residual orientation term is the cubic invariant $G_3 = L^{\otimes 3} T_q$. On the centered balanced submodel, the leading orientation derivative is proportional to $\Delta_{\mathcal{L}} s^3 (\Omega \cdot T_q)$; the simplex-orbit differential Equation (26) makes this term nonzero for every nonzero quotient orientation tangent whenever $\Delta_{\mathcal{L}} > 0$. This gives Equation (54).

The Gaussian derivative envelope, the composed-remainder bound, and the collapsed moment operator yield the uniform all-pair statistical chords in Equations (51)–(52). The simplex cubic stabilizer and polar decomposition yield the physical quotient inverse in Equation (53). Finally, the point-adapted tangent geometry and quadratic-mean differentiability give

$$\|DF_p(\theta)[h]\|^2 \asymp |\dot{p}|^2 + |\dot{\nu}|^2 + \|\dot{b}\|^2 + \|\dot{P}_a\|_F^2 + s^2 \|S\|_F^2 + s^6 \|\Omega\|_F^2,$$

with the p term omitted for a fixed known law. Profiling over nuisance tangents therefore gives Equation (55), and the centered balanced submodel gives Equation (56). $\square$

1.2 Confidence correspondence and physical-resolution upper bound

1.2.1 Explicit median-of-means calibration region

We finish with one fully explicit construction satisfying the calibration coverage interface in Equation (38). Use fixed orthonormal bases of $\text{Sym}^2(\mathbb{R}^q)$ and $\text{Sym}^4(\mathbb{R}^q)$, of dimensions

$$d_2 = \binom{q+1}{2}, \quad d_4 = \binom{q+3}{4}. \quad (46)$$

Let

$$X_2 = \text{symvec}_2(Y^{\otimes 2}), \quad X_4 = \text{symvec}_4(Y^{\otimes 4}).$$

Define

$$\Lambda = \nu_+ + n\lambda_{\max},$$

where $\lambda_{\max} = \max_k \lambda_k$ for the fixed known law and $\lambda_{\max} = 1$ for Bernoulli–Gaussian coding. Every conditional covariance is bounded by ΛI_q , hence

$$v_2^2 := \sup_{\theta} \mathbb{E}_{\theta} \|Y\|^4 \leq \Lambda^2 q(q+2), \quad (47)$$

$$v_4^2 := \sup_{\theta} \mathbb{E}_{\theta} \|Y\|^8 \leq \Lambda^4 q(q+2)(q+4)(q+6). \quad (48)$$

Choose $\alpha_2, \alpha_4 > 0$ with $\alpha_2 + \alpha_4 \leq \alpha_D$ and set, for $k \in \{2, 4\}$,

$$B_k = \left\lceil 8 \log \frac{2d_k}{\alpha_k} \right\rceil, \quad m_k = \left\lfloor \frac{N}{B_k} \right\rfloor. \quad (49)$$

When $m_k \geq 1$, split the first $m_k B_k$ observations into deterministic blocks, compute every coordinate block mean, and take its lower median under a fixed tie rule. The scalar median-of-means inequality plus a coordinate union bound gives

$$\mathbb{P}_{\theta}^N \left\{ \|\widehat{M}_2 - M_2\|_F \leq \epsilon_{2,N} \right\} \geq 1 - \alpha_2, \quad \epsilon_{2,N} = 2v_2 \sqrt{\frac{d_2}{m_2}}, \quad (50)$$

$$\mathbb{P}_{\theta}^N \left\{ \|\widehat{M}_4 - M_4\|_F \leq \epsilon_{4,N}^{\text{raw}} \right\} \geq 1 - \alpha_4, \quad \epsilon_{4,N}^{\text{raw}} = 2v_4 \sqrt{\frac{d_4}{m_4}}. \quad (51)$$

Because normalized symmetrization is an orthogonal projection,

$$\|A \odot B\|_F \leq \|A\|_F \|B\|_F.$$

Also

$$\sup_{\theta} \|M_2(\theta)\|_F \leq B_2^* := \sqrt{q} \Lambda.$$

Therefore

$$\|\widehat{K}_4 - K_4\|_F \leq \epsilon_{K,N} := \epsilon_{4,N}^{\text{raw}} + 3\epsilon_{2,N} (2B_2^* + \epsilon_{2,N}). \quad (52)$$

Thus Equation (38) holds with probability at least $1 - \alpha_D$, uniformly over the fixed-dimensional class. For fixed confidence level and fixed q , both certified radii are $O(N^{-1/2})$ once

$$N \geq N_{\text{MoM}} := \max(B_2, B_4).$$

Below this threshold, the full compact calibration class is retained.

This construction is only one valid realization of the modular calibration coverage interface; any sharper estimator may replace it after its own uniform coverage radii are established.

1.2.2 Measurability of the confidence correspondence

Let $\widetilde{\mathcal{D}}_{q,s}$ be the closed, canonically signed labelled dictionary shell, and define the compact quotient

$$\mathcal{M}_{q,r,s}^p = \frac{[p_-, p_+] \times [\nu_-, \nu_+] \times \widetilde{\mathcal{D}}_{q,s} \times \left(\binom{[q]}{r} \sqcup \{\partial\} \right)}{\mathfrak{S}_q}. \quad (53)$$

The finite group acts simultaneously on coherent-atom labels and supports. No freeness assumption is needed.

For $m = [p, \nu, D, S]$, define

$$L(z, m) = \begin{cases} \min_{x_j \in [\beta_-, \beta_+], j \in S} \left\| z - \sum_{j \in S} x_j d_j \right\|_2, & S \neq \partial, \\ \min_{\gamma \in \Gamma_A} \|z - \gamma a\|_2, & S = \partial. \end{cases} \quad (54)$$

Compact minimization shows that L is jointly continuous on $\mathbb{R}^q \times \mathcal{M}_{q,r,s}^p$. The physical target map ϑ is likewise continuous.

For data $\omega = (y, z^T)$, define

$$\mathcal{R}(\omega) = \left\{ m \in \mathcal{M}_{q,r,s}^p : \pi_{\text{tr}}(m) \in \widehat{\mathcal{K}}_{q,s}^p(y), \quad L(\bar{z}, m) \leq \tau_{T,q} \right\}. \quad (55)$$

Its graph is Borel and its sections are compact. The Arsenin–Kunugui projection theorem therefore makes all open-set hit events Borel, including the nonempty-profile event. Consequently

$$\widehat{\mathfrak{C}} = \vartheta\{\mathcal{R}(\omega)\}$$

on the nonempty event, patched by the fixed admissible singleton on its complement, defines a Borel measurable compact-valued correspondence. The empty-profile indicator is Borel measurable as well.

The target-membership relation is closed in $\mathcal{K}(\mathfrak{V}_q) \times \mathfrak{V}_q$, so the conditional deployment-coverage kernel is Borel in the calibration sample. This justifies the conditional and marginal coverage statements in Theorem 3.3.

1.2.3 Cross-dictionary separation constants

Let

$$c_u = \sqrt{1 - C_0^2 s_0^2}, \quad \lambda_V = \kappa_- \sqrt{\frac{q}{q-1}}.$$

For $h \in \mathbb{R}^q$, write

$$A = \mathbf{1}^\top h, \quad h_0 = h - \frac{A}{q} \mathbf{1}.$$

Because the simplex frame is tight,

$$\|Vh_0\| = \sqrt{\frac{q}{q-1}} \|h_0\|.$$

Using $\|b\| \leq C_0 s$ and $\sigma_{\min}(L) \geq \kappa_- s$ gives a fixed constant $c_D > 0$ such that

$$\left\| \sum_{j=1}^q h_j d_j \right\|_2 \geq c_D s \|h\|_2. \quad (56)$$

Hence two distinct size- r supports satisfy

$$\inf_{x, x'} \|\mu(D, S, x) - \mu(D, S', x')\|_2 \geq \sqrt{2} c_D \beta_- s. \quad (57)$$

Transport across dictionary distance ρ yields

$$m_S(\rho) = \left[\sqrt{2} c_D \beta_- s - C_S r \beta_+ \rho \right]_+, \quad (58)$$

which establishes Equation (58).

For coherent-group versus separated-atom separation, use

$$h_a = \frac{u - e_1}{\sqrt{2}}.$$

The separated-atom neighborhood implies

$$|h_a^\top a| \leq C_a s_0,$$

while every coherent-group mean satisfies

$$h_a^\top \mu(D, S, x) \geq \frac{r}{\sqrt{2}} \{\beta_- c_u - \beta_+ C_0 s_0\}.$$

After shrinking s_0 if needed,

$$g_G^0 := \frac{r}{\sqrt{2}} \{\beta_- c_u - \beta_+ C_0 s_0\} - \gamma_+ C_a s_0 > 0. \quad (59)$$

Cross-dictionary transport then gives

$$m_G(\rho) = [g_G^0 - C_G(r\beta_+ + \gamma_+)\rho]_+, \quad (60)$$

which establishes Equation (59).

Finally, any two retained deployment candidates have fitted means within $2\tau_{T,q}$. Applying the two margins above to all candidate-pair types yields

$$\text{diam}_{\text{pr}}(\widehat{\mathfrak{C}}) \leq C\rho + Cs\mathbf{1}\{2\tau_{T,q} \geq m_S(\rho)\} + C\mathbf{1}\{2\tau_{T,q} \geq m_G(\rho)\}, \quad (61)$$

which gives the deterministic diameter bound used in Theorem 3.3.

1.2.4 Proof of truth retention, coverage, and the physical-resolution upper bound

Proof of Proposition 2.3 and Theorem 3.3. The explicit median-of-means construction above establishes Equation (38); below its fixed block threshold the full compact calibration class is retained. The measurability results above show that the joint feasible set has a Borel graph with compact sections and that $\widehat{\mathfrak{C}}_{q,r,s}^p$ can be chosen as a measurable compact-valued correspondence. This proves the structural part of Proposition 2.3.

Let

$$E_D = \left\{ (p_\star, \nu_\star, [D_\star]) \in \widehat{\mathcal{K}}_{q,s}^p \right\}.$$

By Equation (38), $\mathbb{P}(E_D) \geq 1 - \alpha_D$. Conditional on any calibration realization in E_D ,

$$\|\bar{Z} - \mu_\star\|_2 \leq \tau_{T,q}$$

has probability at least $1 - \alpha_T$, uniformly over the allowed deployment noise levels, and retains the true support and coefficient witness. Hence

$$\vartheta_\star \in \widehat{\mathfrak{C}}_{q,r,s}^p, \quad F_{\text{empty}} = 0$$

on the intersection of the two truth-retention events. This proves Proposition 2.3, Equation (60), and, by independence, Equation (61).

For two members of the same nonempty calibration region, Equations (30)–(36) imply

$$|p - p'| + \delta_F\{(D, \nu), (D', \nu')\} \leq C\epsilon_N. \quad (62)$$

Combining this with the physical quotient inverse gives Equation (62); the certified $\epsilon_N = O(N^{-1/2})$ rate gives Equation (63).

It remains to control coexistence of retained deployment explanations. The cross-dictionary separation lemmas above provide the support margin $m_S(\rho)$ and coherent-group/separated margin $m_G(\rho)$. Any two retained candidates have fitted means within $2\tau_{T,q}$ of each other. Therefore candidates with different deployment alternatives cannot coexist when $2\tau_{T,q} < m_G(\rho)$, and distinct coherent-group supports cannot coexist when $2\tau_{T,q} < m_S(\rho)$. When the support margin is positive, the quotient geometry also gives a unique matching of nearby coherent atoms; candidates on the same support branch then differ physically by $O(\rho)$. Applying these cases with $\rho = \rho_N(s)$ yields Equation (64) and completes the proof. $\square$

1.3 Minimax physical resolution

1.3.1 Proof of the minimax lower bound and resolved optimality

Proof of Theorem 3.5 and Corollaries 3.6–3.7. For two joint parameters ξ_0, ξ_1 , let $\vartheta(\xi_i) = (m_i, A_i)$ and set

$$d = d_H^{\text{pr}}(A_0, A_1).$$

If a random compact correspondence has coverage at least $1 - \alpha$ at both endpoints, then

$$\mathbb{E}_{\xi_0} \text{diam}_{\text{pr}}(\widehat{C}) \geq d \left\{ 1 - 2\alpha - \text{TV}(\mathbb{P}_{\xi_0}^{N,T}, \mathbb{P}_{\xi_1}^{N,T}) \right\}_+. \quad (63)$$

Indeed, under $\mathbb{P}_{\xi_0}$ the two target-inclusion events intersect with probability at least the factor in braces, and on this intersection the physical diameter is at least d .

We use three two-point submodels.

Coherent-group versus separated alternative. On the balanced submodel, choose $x = \beta_- \mathbf{1}_S$, $\sigma = \sigma_+$, and $\gamma_\star = a_0^\top \mu_F \in \text{int}(\Gamma_A)$. The calibration laws are identical and

$$\text{KL}_{N,T} = \frac{T}{2\sigma_+^2} \|(I - P_{a_0})\mu_F\|_2^2 \leq CI_G^{(r)}. \quad (64)$$

The physical-support components remain separated by a fixed positive amount. Equation (63) therefore yields the order-one lower term.

Wrong-support pair. Keep the dictionary fixed, choose two supports sharing $r - 1$ atoms, and use the common coefficient β_- . If i, j are the exchanged atoms,

$$\|\mu_S - \mu_{S'}\|_2^2 = 2\beta_-^2 \lambda_\star^2 s^2 \frac{q}{q-1}, \quad (65)$$

so the joint KL is $O(I_S)$, while the projective Hausdorff separation of the physical supports is $O(s)$ and bounded below by a fixed multiple of s . This gives the order- s lower term.

Compensated orientation pair. At the centered balanced submodel with equal active coefficients, let

$$w_S = \sum_{j \in S} v_j.$$

Choose a nonzero skew generator Ω that fixes w_S while moving at least one active vertex, and normalize it so that $\max_{j \in S} \|\Omega v_j\| = 1$. For sufficiently small fixed h_0 , the same-label matching remains optimal along $R \exp(h\Omega)$ and

$$cs|h| \leq d_{\text{phys}}\{\vartheta(D_0, S), \vartheta(D_h, S)\} \leq Cs|h|, \quad |h| \leq h_0. \quad (66)$$

The complete deployment law is identical along this compensated path, while the calibration KL is at most CNs^6h^2 by Equation (51). Taking

$$|h| \asymp \min \left\{ h_0, (\sqrt{N} s^3)^{-1} \right\}$$

and applying Equation (63) yields

$$s \wedge \frac{1}{\sqrt{N} s^2}.$$

Together, the three pairs prove Equation (71).

For the matching upper bound, set

$$R_D(s, N) = s \wedge (\sqrt{N} s^2)^{-1}.$$

Equation (63) gives $\rho_N \leq C_\rho R_D$. Choose the fixed shell radius and the resolution thresholds so that the coherent-group/separated indicator in Equation (64) vanishes whenever $I_G^{(r)}$ exceeds its upper threshold. If R_D is a fixed fraction of s , the support term is already $O(R_D)$; if R_D is smaller, then $\rho_N = o(s)$, the support margin is positive, and the upper threshold on I_S removes the support indicator. Thus the right-hand side of Equation (64) is $O(R_D)$, proving Equation (72).

For Corollary 3.6, set the dictionary uncertainty to zero. Then $I_S \rightarrow \infty$ separates the fine supports and $I_G^{(r)} \rightarrow \infty$ excludes the separated alternative, so the physical diameter tends to zero. In the learned-dictionary problem, the compensated orientation pair retains an order- s physical displacement whenever Ns^6 remains bounded.

For Corollary 3.7, the same compensated path already fixes the coherent group, support, equal coefficients, separated atom, coefficient law, noise levels, and all nonorientation nuisance parameters. Hence the lower bound survives after these quantities are revealed by an oracle. $\square$

1.4 Deployment-assisted orientation information

1.4.1 Two-atom physical target secant

For $S = \{1, 2\}$ on the balanced submodel, write

$$e = v_1 + v_2, \quad f = v_1 - v_2.$$

Choose $g \perp \text{span}\{e, f\}$ with $\|g\| = \|f\|$, and a skew generator Ω such that

$$\Omega e = 0, \quad \exp(h\Omega)f = \cos(h)f + \sin(h)g.$$

Then

$$\Omega v_1 = \frac{1}{2}g, \quad \Omega v_2 = -\frac{1}{2}g.$$

For the coherent atoms along this orbit, a sufficiently short compact chart satisfies

$$c_1 s|h - h'| \leq d_{\text{pr}}([d_j(h)], [d_j(h')]) \leq C_1 s|h - h'|, \quad j = 1, 2. \quad (67)$$

Choosing the chart short enough that cross-label matches remain separated makes the same-label matching Hausdorff-optimal, so

$$c_V s|h - h'| \leq d_H^{\text{pr}}(\{[d_1(h)], [d_2(h)]\}, \{[d_1(h')], [d_2(h')]\}) \leq C_V s|h - h'|. \quad (68)$$

For coefficients

$$x_1 = \bar{\beta} + \frac{d}{2}, \quad x_2 = \bar{\beta} - \frac{d}{2},$$

exact coefficient profiling yields

$$\inf_{\bar{\beta}', d' \in \mathbb{R}} \|\mu_{h, \bar{\beta}, d} - \mu_{h', \bar{\beta}', d'}\|_2 = \frac{\lambda_\star s |d| \|f\|_2}{2} |\sin(h - h')|. \quad (69)$$

On any compact sign-preserving coefficient set with $|d| \asymp |d_0| > 0$, this verifies the two-sided deployment secant used in Theorem 3.9. On the separate exact slice $d = 0$, the deployment mean is invariant along the orientation orbit.

1.4.2 Proof of the deployment-information results

Proof of Theorem 3.9 and Corollary 3.10. Let

$$w_{S,x} = \sum_{j \in S} x_j v_j, \quad \mathcal{V}_S^0 = \left\{ \sum_{j \in S} c_j v_j : \sum_{j \in S} c_j = 0 \right\}.$$

At the centered balanced submodel, an orientation tangent changes the deployment mean by

$$\dot{\mu}_R = \lambda_\star s R \Omega w_{S,x}.$$

Coefficient perturbations with zero sum span $\lambda_\star s R \mathcal{V}_S^0$, while a nonzero coefficient sum creates an axial component and cannot cancel a pure order- s transverse orientation change. Orthogonal projection therefore gives Equation (74), and the Gaussian likelihood gives Equation (75).

For the restricted orientation orbit, two retained parameters separated by $h = \phi - \phi'$ must satisfy

$$s^3 |h| \lesssim N^{-1/2}$$

from the calibration profile and

$$s \chi_s |h| \lesssim \sigma T^{-1/2}$$

from the deployment profile. The target secant therefore gives the upper rate in Equation (79). For the lower bound choose

$$|h| \asymp \min \left\{ h_0, (N s^6 + T \chi_s^2 s^2 / \sigma^2)^{-1/2} \right\}.$$

Calibration and deployment KL divergences add by independence, while the target separation is at least $cs|h|$. The two-point diameter bound in Equation (63) gives the matching lower rate.

For $r = 2$, the technical secant calculation above gives

$$\inf_{\bar{\beta}', d' \in \mathbb{R}} \|\mu_{\phi, \bar{\beta}, d} - \mu_{\phi', \bar{\beta}', d'}\|_2 = \frac{\lambda_\star s |d| \|v_1 - v_2\|_2}{2} |\sin(\phi - \phi')|.$$

On the declared sign-preserving coefficient set, $|d| \asymp |d_0|$, which verifies the two-sided deployment secant and yields Equation (81). When $d = 0$, the deployment mean is exactly invariant along the orientation path, so deployment replication provides no orientation information. $\square$

2 Finite-Bank Implementation and Numerical Details

This section collects the numerical and implementation details omitted from the main text. It has two distinct purposes. First, we document the theory-guided calculation used to illustrate the local s^6 orientation information and the deployment task-symmetry mechanism. Second, we specify the finite-bank evaluator, certificate logic, and experimental protocols used in the global and four-region AEB studies.

These calculations do not numerically implement the continuous confidence correspondence of Section 2, and the finite banks are not assumed to approximate or outer-cover the continuous parameter space.

2.1 Theory-guided numerical diagnostics

2.1.1 Balanced model and exact calibration mixture

Let $v_1, \dots, v_4 \in \mathbb{R}^3$ be the tetrahedral vertices

$$3^{-1/2}(1, 1, 1), \quad 3^{-1/2}(1, -1, -1), \quad 3^{-1/2}(-1, 1, -1), \quad 3^{-1/2}(-1, -1, 1).$$

With $u = e_4$, define the coherent atoms along the orientation path by

$$d_j(s, \phi) = \sqrt{1 - s^2} u + s R_\phi v_j, \quad (70)$$

where R_ϕ rotates about $v_1 + v_2$. Thus $R_\phi(v_1 + v_2) = v_1 + v_2$, and the deployment mean on $S = \{1, 2\}$ with equal coefficients is invariant. The separated atom is $(u + e_1)/\sqrt{2}$.

We use

$$s \in \{0.045, 0.055, 0.067, 0.082, 0.100, 0.122\}, \quad p = 0.2, \quad \nu = 0.5,$$

unit active-code variance, and path parameter $h = 0.25$. The skew generator is normalized by $\max_{j \in S} \|\Omega v_j\| = 1$.

For a support mask $a \in \{0, 1\}^5$, let

$$w_a = p^{|a|} (1 - p)^{5 - |a|}, \quad \Sigma_a(\phi) = \nu I_4 + D_\phi \text{diag}(a) D_\phi^\top.$$

The one-observation calibration density is evaluated as the exact 32-component Gaussian mixture

$$f_{s, \phi}(y) = \sum_{a \in \{0, 1\}^5} w_a \varphi_4\{y; 0, \Sigma_a(\phi)\}. \quad (71)$$

Hence this diagnostic uses neither a dictionary-learning optimizer nor a numerical approximation to the continuous confidence region.

2.1.2 Stable divergence calculation

Let

$$P = f_{s, 0}, \quad Q = f_{s, \phi}, \quad M = \frac{P + Q}{2}, \quad L = \log \frac{dP}{dQ}.$$

To avoid cancellation in the near-coherent regime, we use the nonnegative midpoint identities

$$J(P, Q) = 2 \mathbb{E}_M \{L \tanh(L/2)\}, \quad (72)$$

$$1 - A(P, Q) = \mathbb{E}_M \left[\frac{\tanh^2(L/2)}{1 + \text{sech}(L/2)} \right], \quad (73)$$

where J is Jeffreys divergence and $A(P, Q) = \int \sqrt{dP dQ}$ is Hellinger affinity.

At each value of s , 128 common-random-number batches of size 8192 give 2^{20} midpoint observations. Paired resampling across the six scales produces the reported slope interval. As an independent deterministic check, tensor Gauss–Hermite rules of orders 10 and 14 integrate every Gaussian component in Equation (71); the order-14 all-scale fit has exponent 5.9342 and $R^2 = 0.99999$.

For N independent calibration observations,

$$A(P^{\otimes N}, Q^{\otimes N}) = A(P, Q)^N. \quad (74)$$

For two Gaussian deployment means with common covariance $\sigma^2 I/T$, the affinity is

$$\exp \left\{ -\frac{T \|\mu_0 - \mu_1\|_2^2}{8\sigma^2} \right\}. \quad (75)$$

These identities are used only to visualize the separate calibration, support-discrimination, and deployment-model information scales.

At each fixed separation scale, 128 common-random-number batches of 8192 midpoint observations—4096 from each endpoint mixture—estimate the Jeffreys divergence, for 2^{20} observations per scale. The same batch index is paired across all six fixed scales. For each of 2000 bootstrap replicates (seed 2026072202), 128 batch indices are sampled with replacement, the batch estimates are averaged within each scale, and the ordinary least-squares slope of log mean divergence against log separation scale is recomputed. The reported central 95% range is the 2.5th and 97.5th percentiles of those slopes. It is a Monte Carlo stability diagnostic conditional on the fixed mixture model and grid, not uncertainty across independent datasets. The underlying midpoint Monte Carlo seed is 2026072201; the per-scale error bars are separate Student- t intervals over the 128 batch means.

Table 1: Theory-guided numerical diagnostics. Monte Carlo intervals use the 128 paired batches; quadrature and deployment checks are deterministic.

Diagnostic	Result	Interpretation
Jeffreys s -exponent	5.9355; central 95% paired-batch MC range [5.9249, 5.9468]	sixth-order scaling; numerical stability
Jeffreys log-log R^2	0.99999	stable scaling
Order-14 quadrature exponent	5.9342	independent check
Maximum cross- s affinity spread	0.0243	approximate collapse
Median transition-band spread	0.0167	finite- s drift
Prespecified spread tolerance	0.015	not met
Equal-coefficient deployment residual	6.82×10^{-16}	invariant to numerical precision
Contrast-formula relative error	7.01×10^{-15}	analytical agreement

Across-scale collapse diagnostic

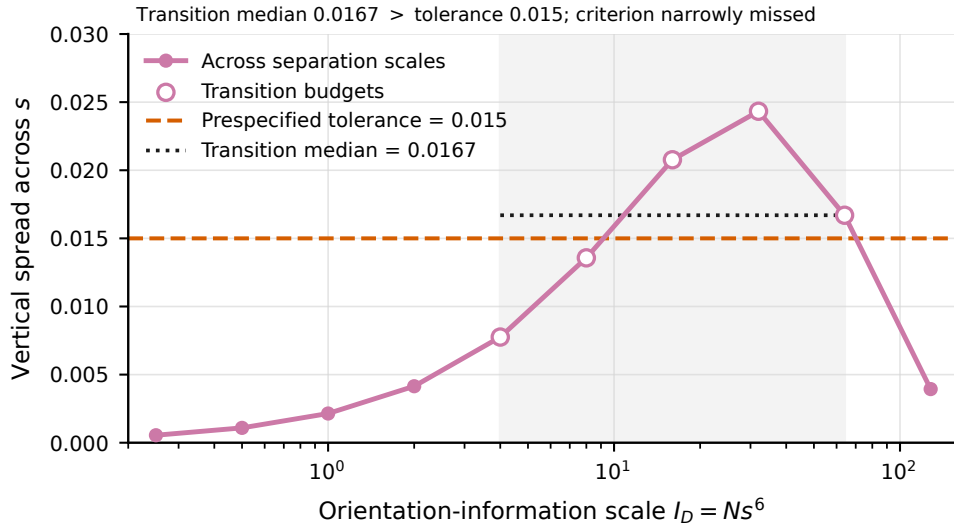


Figure 1: Across-scale orientation-information collapse diagnostic underlying Figure 2. At each stored $I_D = Ns^6$, the plotted vertical spread is the maximum minus the minimum product affinity across the saved separation scales. Over the prespecified transition budgets $I_D \in \{4, 8, 16, 32, 64\}$, the median spread is 0.0167, which narrowly exceeds the prespecified tolerance 0.015. The criterion is therefore recorded as missed and is not used as evidence of numerical coverage for the continuous confidence correspondence.

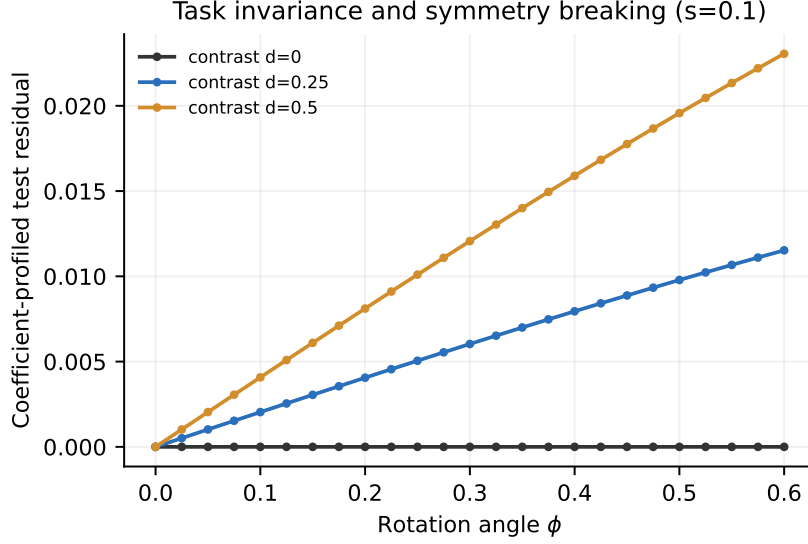


Figure 2: Coefficient-profiled deployment residual at $s = 0.1$. Equal coefficients are invariant to numerical precision, whereas nonzero coefficient contrasts follow the closed form in Equation (80).

2.1.3 Deployment task control and robustness checks

For contrasts $d \in \{0.25, 0.5\}$, the numerical least-squares residual is compared with

$$\frac{s|d|\|v_1 - v_2\|_2}{2} |\sin \phi|.$$

The largest relative discrepancy over the nonzero displayed cells is 7.01×10^{-15} . Optimal projective matching preserves the same coherent-atom labels throughout the displayed path.

Additional robustness checks were performed after the primary calculation. The fitted h -exponent is 1.943 with $R^2 = 0.99983$. Across the nine cells

$$p \in \{0.1, 0.2, 0.35\}, \quad \nu \in \{0.3, 0.5, 1.0\},$$

every log-log R^2 exceeds 0.9999. The smallest fitted s -exponent is 5.759, occurring at the smallest activation probability and noise level. These calculations are reported as robustness diagnostics, not as a new fitted theory.

2.2 Finite-bank evaluator and certificate implementation

Let

$$\mathcal{B} = \{e_1, \dots, e_M\}$$

be a finite bank fixed before held-out candidate evaluation. For candidate e , let $\ell_t(e)$ denote its log likelihood-ratio e-value at checkpoint t , and define

$$\tau_\alpha = -\log \alpha, \quad m(e) = \max_t \ell_t(e) - \tau_\alpha. \quad (76)$$

After the required numerical evaluation, $m(e) \leq 0$ is classified ADMISSIBLE and $m(e) > 0$ is classified REJECTED. A queried candidate whose sign cannot be certified is INDETERMINATE. Unqueried and indeterminate candidates are never treated as rejected.

After k logical queries, AEB maintains

$$\begin{aligned} L_k &= \{e \in \mathcal{B} : e \text{ has been queried and is admissible}\}, \\ U_k &= \{e \in \mathcal{B} : e \text{ has not been rejected}\}. \end{aligned} \quad (77)$$

If $\mathcal{A}_\mathcal{B}$ is the exhaustive profile obtained by applying the same candidate rule to the full bank, then the implementation maintains

$$L_k \subseteq \mathcal{A}_\mathcal{B} \subseteq U_k. \quad (78)$$

Fix the requested report family $\mathcal{G}$ before held-out evaluation, and let $\mathcal{O}_{\mathcal{G}}$ denote its allowed reports. The implementation uses the deterministic certificate rule

$$\text{Cert}_{\mathcal{G}}(L, U) \in \mathcal{O}_{\mathcal{G}} \cup \{\perp\},$$

where $\perp$ means that the requested report is not yet complete. A truth-level field is emitted only when its pointwise predicate holds for every candidate in U , while an ambiguity-level field is emitted only when its prespecified opposing admissible witnesses are present in L . Thus a substantive report is emitted only when $\text{Cert}_{\mathcal{G}}(L, U) \neq \perp$. Adaptive candidate ordering uses the predeclared report-family-specific priority function $\pi_{\mathcal{G}}(e; L, U)$ and its deterministic tie-breaking rule. The detailed global and four-region proposal and witness orders below are specific implementations of this generic rule.

Empty profile remains a separate terminal state and is not an output of $\text{Cert}_{\mathcal{G}}$. A substantive exhaustive-profile map additionally requires the exhaustive profile to be nonempty. If all candidates have been classified and none is admissible, the interface returns an empty-profile status and a null physical map. A query-limited state with no admissible witness is not interpreted as an empty profile; it leads to ABSTAIN unless an independently valid truth-level field is already certified.

2.2.1 Conditional candidate retention

Let $\mathcal{H}_{\text{prop}}$ contain the data split, all proposal-stage observations, and the proposal and policy objects constructed from them. Conditional on $\mathcal{H}_{\text{prop}}$, the proposal density is fixed before any held-out evaluation observation is used. For the generic one-stream evaluator, let $W = (W_1, \dots, W_m)$ denote the held-out evaluation sample. For a fully specified on-bank candidate e , the evaluator uses

$$E_t(e) = \prod_{i \leq t} \frac{f_e^{\sim}(W_i)}{f_e(W_i)}. \quad (79)$$

Under $\mathbb{P}_e$,

$$\mathbb{E}_e \left[\frac{f_e^{\sim}(W_t)}{f_e(W_t)} \middle| \mathcal{F}_{t-1}, \mathcal{H}_{\text{prop}} \right] = 1,$$

so $E_t(e)$ is a nonnegative conditional martingale. The proposal density may lie outside the finite candidate bank; only its conditional independence from the held-out evaluation sample is used.

The application-specific evaluator below specializes the generic held-out sample W into independent calibration and deployment streams, denoted Y and Z . The four-region experiment uses these two evaluation streams. For candidate $e = (g, S)$,

$$E_k(e) = \prod_{i \leq n_k} \frac{f_g^C(Y_i)}{f_g^C(Y_i)} \prod_{j \leq t_k} \frac{f_e^D(Z_j)}{f_e^D(Z_j)} \quad (80)$$

along the deterministic checkpoint path

$$(n_k, t_k) \in \{(128, 0), (256, 0), (512, 0), (1024, 0), (2048, 0), (2253, 116)\}. \quad (81)$$

Conditional Ville's inequality therefore gives candidate-wise rejection probability at most α when rejection requires a strict crossing of $1/\alpha$. This is a guarantee for the represented true candidate, not simultaneous nonrejection of all candidates.

The implementation evaluates the likelihood-ratio scores in binary64 and re-evaluates designated near-threshold or certificate-critical candidates at 90-decimal precision. This high-precision check is not directed interval arithmetic. The formal probability statement is therefore the exact-real candidate-wise result above; the numerical implementation is documented to make its finite candidate classifications reproducible.

2.3 Global finite-bank evaluation protocol

The global study uses

$$q = 4, \quad n = 5, \quad \phi_{\text{truth}} = 0, \quad p_{\text{truth}} = 0.20,$$

32 calibration-mixture components and active support $\{0, 1\}$. Each dataset is split into proposal and held-out evaluation fractions 0.45 and 0.55. The global study fixes its requested report family, certificate rule, priority function, and deterministic tie-breaking order before held-out evaluation.

The proposal orientation $\widehat{\phi}_A$ is retained as a fixed reference in the global diameter calculation. If S denotes the exhaustive set of surviving bank orientations, define

$$\begin{aligned} W_k^+ &= \{\widehat{\phi}_A\} \cup \{\phi_j : \phi_j \text{ has a queried admissible nuisance witness}\}, \\ P_k^+ &= \{\widehat{\phi}_A\} \cup \{\phi_j : \phi_j \text{ has not been eliminated}\}, \\ S^+ &= \{\widehat{\phi}_A\} \cup S. \end{aligned}$$

Correct candidate classifications give

$$W_k^+ \subseteq S^+ \subseteq P_k^+, \quad \text{diam}(W_k^+) \leq \text{diam}(S^+) \leq \text{diam}(P_k^+).$$

Thus this experiment evaluates a proposal-referenced global physical-resolution summary; it does not use the regional support predicates of the four-region experiment.

Table 2: Global finite-bank evaluation conditions.

Condition	N	s	ν_{truth}	Bank size	Cases
Low information	4096	0.35	1.20	1025	6
Intermediate information	65536	0.50	0.90	1025	6
High information	131072	0.50	0.80	369	6

The data seeds are 2026080111–2026080116, 2026080121–2026080126, and 2026080131–2026080136 for the three conditions, respectively. The corresponding split seeds are obtained by replacing the leading 2 with 9.

Each dataset is evaluated under three independently specified reporting profiles:

	α	$\delta_f/d_{\text{shell}}$	$\delta_s/d_{\text{shell}}$
RISK-CONSERVATIVE	0.025	0.40	0.50
BALANCED	0.077	0.35	0.60
RESOLUTION-FAVORING	0.150	0.25	0.40

giving 54 controller traces from 18 independent datasets. Results are recorded at query-budget fractions

$$\{0.10, 0.20, 0.35, 0.50, 0.75, 1.00\}.$$

A logical query is counted whenever the controller requests a candidate classification, even if a numerical score is already cached.

After every controller trace has been fixed, the same finite bank is evaluated exhaustively using the same candidate-classification rule. The finite-bank reference therefore does not influence the adaptive query order. Near-threshold candidates and candidates affecting terminal or elimination certificates receive the high-precision re-evaluation described above.

The principal denominators in the main-text summary are 54 controller traces, 34 exhaustive-bank ambiguity traces, seven exhaustive-bank fine traces, and 20 exhaustive-bank nonambiguous traces. Full-budget categorical agreement is evaluated over all 54 traces. The 54 traces arise from three reporting profiles applied to 18 independent datasets and must not be interpreted as 54 independent datasets.

The median queried bank fraction is 0.209 at both budget 0.50 and budget 0.75. This is the median of normalized trace-specific query fractions, not raw query counts. Thirty-four terminal fractions are already at or below 0.50; the 27th and 28th order statistics are 201/1025 and 227/1025, whose average is $214/1025 = 0.208780\dots$. These middle observations are therefore fixed before either displayed cap. Raising the cap changes the upper tail and fine recovery rather than the median stopping fraction.

2.4 Four-region application protocol

The four-region response library contains 12 atoms, 72 dictionary states, and the support patterns AB, ABC, and ABD, giving

$$M = 216$$

complete candidate explanations. The four-region study fixes its requested report family, certificate rule, priority function, and deterministic tie-breaking order before held-out evaluation. The fixed design parameters are

$$\alpha = 0.077, \quad h = 0.085, \quad \tau_A = 1, \quad \tau_B = 0.8, \quad \tau_C = 0.1, \quad \tau_D = 1.00.$$

For candidate $e = (g, S)$,

$$\Sigma_e = \nu_D I + \sum_{R \in S(e)} \tau_R^2 a_R(g) a_R(g)^\top. \quad (82)$$

The represented regional strength is

$$\lambda_R(e) = \tau_R \mathbf{1}\{R \in S(e)\}.$$

Accordingly, a reported absence statement excludes the optional-region-present candidate family at the represented scale τ_R . It does not assert absence over a continuum of amplitudes or arbitrarily weak off-bank sources.

Each dataset contains a 4096×16 calibration array and a 192×16 deployment array. The proposal/evaluation partition contains 1843/2253 calibration observations and 76/116 deployment observations. The six evaluation checkpoints are exactly those in Equation (81).

The 15 fresh datasets consist of six persistent-only cases with seeds 2026091001–2026091006, six weak-C cases with seeds 2026091011–2026091016, and three D-present controls with seeds 2026091021–2026091023.

The regional policy alternates between actions that can establish opposing support witnesses and actions that can eliminate candidates preventing a regional certificate. Its fixed proposal order is the deployment proposal dictionary, followed by the distinct calibration proposal dictionary, followed by candidate ID. For the weak-C ambiguity objective, the fixed priority order is: (i) obtain a C-absent witness, (ii) obtain a C-present witness, (iii) eliminate an alternative A location, and (iv) eliminate the D-present family. After each query, the regional rule $\text{Cert}_G(L, U)$ is recomputed. The primary query cap is

$$162/216.$$

The 108/216 state reported in diagnostics is a prefix of the same controller trace, not a separate run.

The double-precision rejection threshold is

$$-\log(0.077) = 2.563949857.$$

Provisionally admissible, near-threshold, and certificate-critical candidates receive the high-precision re-evaluation used by the exhaustive finite-bank reference. The exhaustive reference scores all 216 candidates and projects only the admissible subset to the regional physical map.

Fourteen of the 15 datasets yield nonempty exhaustive profiles. The completed-profile denominators used in the main text are: 11 eligible A-fine cases, five B group/sector cases, five C-ambiguity cases, and ten represented-scale D-absence cases. The point-valued plug-in comparison uses the same five eligible weak-C cases for both region B and region C. The three D-present controls provide the denominator for the false-D-absence check.

One weak-C realization produces an empty exhaustive finite profile. At the 162-query online cap its lower set is empty and 54 candidates remain possible, so AEB returns ABSTAIN; only subsequent exhaustive scoring shows that no candidate in the finite bank is admissible. The reported physical map is therefore null. This realization is excluded from truth-relative utility rates based on completed profiles, but its 162 queries remain in the computational summary. It is not interpreted as evidence that all physical regions are absent.

The three D-present controls also reach the primary cap without an admissible lower-set witness, although their exhaustive profiles are nonempty. Their universal regional truth-level fields may be inspected separately, but they are not counted as witnessed global nonabstaining maps. None receives the represented-scale D-absence label.

2.5 Query accounting, reproducibility, and scope

A logical query is one controller-requested candidate evaluation. Cache hits do not remove logical queries. Offline exhaustive finite-bank scoring, high-precision verification, completion checks, and report generation are excluded from the adaptive query count and are not interpreted as deployment latency.

The global and four-region studies use different banks, report functionals, and query policies; their observations and denominators are not pooled. The publication package records the candidate laws, data splits, checkpoint grids, proposal and witness priorities, tie order, report predicates, seeds, raw numerical summaries, and figure data required to reproduce the reported experiments. For the theory-guided calculation, the Monte Carlo seed is 2026072201 and the paired-bootstrap seed is 2026072202.

The numerical evidence has deliberately limited scope. The theory-guided experiment illustrates the finite- s manifestation of the s^6 calibration geometry and deployment task symmetry. The finite-bank experiments evaluate AEB relative to exhaustive evaluation of the same specified candidate banks. They do not establish numerical coverage of the continuous confidence correspondence, completeness of the finite banks relative to the continuous model, off-bank validity, worst-case sublinear query complexity, or transfer to arbitrary real-data applications.